

Refutation of Conjecture 00000004007: the claimed optimal subadditivity inequality for p -variation is false

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13 September 2026

Abstract

We disprove Conjecture 00000004007. The p -variation of a path is the p -th root of the supremum over partitions of the sum of the p -th powers of the increments. Being a p -th root of a supremum, it is homogeneous of degree one, $\|aX\| = |a| \|X\|$. Taking $X = Y \neq 0$ gives $\|X + Y\| = 2 \|X\|$ on the left and $2^{1/q} \|X\|$ on the right, where $q = p/(p - 1)$ is the conjugate exponent. Since $q > 1$ for every finite $p > 1$, we have $2 > 2^{1/q}$ and the conjectured inequality fails. The same conclusion holds for the raw-supremum normalisation (without the root), a fortiori, and for the limit case $p = 1$ ($q = \infty$). The failure is elementary; the inequality cannot be “enlarged” because it is not valid in the first place. The arithmetic core of the obstruction is formalised in core Lean 4.

1 The conjecture

Let $X : [0, 1] \rightarrow V$ be a path with values in a normed vector space V , and let a *partition* π of $[0, 1]$ be a finite increasing sequence

$$\pi : \quad 0 = t_0 < t_1 < \cdots < t_k = 1.$$

The statement under consideration is the following.

Conjecture 1 (00000004007). With the p -variation defined as the p -th root of the supremum over partitions of the sum of the p -th powers of the increments, and with q the conjugate exponent of p , the inequality

$$\|X + Y\| \leq (\|X\|^q + \|Y\|^q)^{1/q} \tag{1}$$

holds, and it cannot be enlarged.

We show that (1) is false, so the conjecture is false.

2 Definition and homogeneity

Definition 2. For $1 \leq p < \infty$ the *p-variation* of a path X is

$$\|X\| := \left(\sup_{\pi} \sum_{i=1}^k \|X(t_i) - X(t_{i-1})\|^p \right)^{1/p}, \quad (2)$$

the supremum being over all partitions $\pi = \{0 = t_0 < \dots < t_k = 1\}$. The conjugate exponent of p is $q := \frac{p}{p-1}$, characterised by $\frac{1}{p} + \frac{1}{q} = 1$; thus $q \in (1, \infty)$ whenever $p \in (1, \infty)$, and $q = \infty$ when $p = 1$.

Lemma 3 (Homogeneity). *For every scalar a and every path X ,*

$$\|aX\| = |a| \|X\|. \quad (3)$$

Proof. The increments of aX are $a(X(t_i) - X(t_{i-1}))$, so for each fixed partition π ,

$$\sum_{i=1}^k \|a(X(t_i) - X(t_{i-1}))\|^p = |a|^p \sum_{i=1}^k \|X(t_i) - X(t_{i-1})\|^p.$$

The factor $|a|^p \geq 0$ is independent of π , and for $c \geq 0$ one has $\sup_{\pi}(cS_{\pi}) = c \sup_{\pi} S_{\pi}$. Hence the p -th powers of the suprema are related by $|a|^p$, and taking the p -th root gives

$$\|aX\| = (|a|^p \|X\|^p)^{1/p} = |a| \|X\|. \quad \square$$

Remark 4. The homogeneity (3) is exactly the property that makes the conjecture fail: it forces both sides of (1) to scale linearly in the path, and the two linear coefficients disagree for $X = Y$.

3 Refutation: the case $X = Y$

Let X be any non-zero path and take $Y := X$. Using Lemma 3, the left-hand side of (1) is

$$\|X + Y\| = \|2X\| = 2 \|X\|,$$

while the right-hand side is

$$(\|X\|^q + \|Y\|^q)^{1/q} = (2\|X\|^q)^{1/q} = 2^{1/q} \|X\|.$$

Since $X \neq 0$ we have $\|X\| > 0$, so (1) at $X = Y$ is equivalent to

$$2 \leq 2^{1/q} \iff 2^q \leq 2. \quad (4)$$

But $q = p/(p-1) > 1$ for every finite $p > 1$, and $2^q > 2$ for every $q > 1$. Therefore (4) is false, and the conjectured inequality (1) fails. It “cannot be enlarged” for the trivial reason that it does not hold at all.

Remark 5 (Elementary nature of the failure). No estimate, tightness analysis, or special path is required. The counterexample is the simplest imaginable one, $X = Y \neq 0$; the failure is a direct consequence of the degree-one homogeneity (3) and of $q > 1$. This is precisely why the stated inequality cannot be correct.

4 Explicit values for a two-point unit-step path

Let X be the two-point path with a single unit step, $X(0) = 0$, $X(1) = 1$; there is only the trivial partition $\{0, 1\}$, so

$$\|X\| = (|1|^p)^{1/p} = 1.$$

For $Y = X$ the path $X + Y$ has a single step of size 2, hence

$$\|X + Y\| = (2^p)^{1/p} = 2,$$

independently of p , while the claimed right-hand side is $(1^q + 1^q)^{1/q} = 2^{1/q}$. The following table lists the values; in every case the left-hand side strictly exceeds the right-hand side.

p	$q = \frac{p}{p-1}$	$\ X\ $	$\ X + Y\ $	RHS = $2^{1/q}$	LHS > RHS
3/2	3	1	2	1.2599...	yes
2	2	1	2	1.4142...	yes
3	3/2	1	2	1.5874...	yes
4	4/3	1	2	1.6818...	yes
10	10/9	1	2	1.8661...	yes

For $p = 2$ the requirement (4) reads $4 \leq 2$, and for $p = 3$ it reads $2^{3/2} \leq 2$, i.e. $8 \leq 4$; both are false. As p grows, $q = p/(p-1) \downarrow 1$ and $2^{1/q} \uparrow 2$ from below, so the margin $2 - 2^{1/q}$ shrinks but never vanishes for finite p : the inequality fails for *every* $p > 1$.

5 The raw-supremum reading

The conjecture specifies the p -th root in the definition (2). If instead one reads $\|\cdot\|$ as the *raw* supremum, without taking the p -th root,

$$N(X) := \sup_{\pi} \sum_{i=1}^k \|X(t_i) - X(t_{i-1})\|^p, \quad (5)$$

then N is homogeneous of degree p , $N(aX) = |a|^p N(X)$. At $X = Y$ with $\|X\|$ denoting the raw quantity (5),

$$N(X + Y) = N(2X) = 2^p N(X),$$

and the claimed right-hand side is

$$(2N(X)^q)^{1/q} = 2^{1/q} N(X),$$

so the inequality would require $2^p \leq 2^{1/q}$. For every $p \geq 1$ and $q > 1$ we have $2^p \geq 2 > 2^{1/q}$, so the raw reading fails a fortiori; for instance $p = 2$ gives 4 on the left against $2^{1/2} = 1.4142\dots$ on the right. Hence the refutation does not depend on which of the two normalisations is intended.

6 The limit case $p = 1$

For $p = 1$ the quantity (2) is the total variation and the conjugate exponent is $q = \infty$, with

$$(\|X\|^q + \|Y\|^q)^{1/q} \xrightarrow{q \rightarrow \infty} \max\{\|X\|, \|Y\|\}.$$

Taking $X = Y \neq 0$ gives

$$\|X + Y\| = 2\|X\| > \|X\| = \max\{\|X\|, \|Y\|\},$$

so (1) fails in this case as well.

7 Conclusion

Conjecture 00000004007 is false. For every $p \in (1, \infty)$ the conjugate exponent satisfies $q > 1$, and the single choice $X = Y \neq 0$, together with the degree-one homogeneity of the p -variation, reduces the claimed inequality to the false numerical statement $2^q \leq 2$. The failure is elementary and occurs for all admissible p ; the raw-supremum reading and the limit case $p = 1$ fail as well. There is no sense in which the inequality can be “enlarged”: it is simply not a true inequality.

8 Formal verification

The arithmetic core of the obstruction is formalised in core Lean 4 (`lean4/Main.lean`; `import Std`, no `Mathlib`, no `sorry`, no `axiom`, no `native_decide`, no $\mathbb{R}$). The formalised statements are:

- `q_eq_two_fails` : $\neg(2^2 \leq 2)$, i.e. $\neg(4 \leq 2)$ — the case $p = 2$ ($q = 2$);
- `q_eq_two` : $2^2 = 4$;
- `q_eq_three_halves_fails` : $\neg(2^3 \leq 2^2)$, i.e. $\neg(8 \leq 4)$ — the case $p = 3$ ($q = 3/2$, since $2^{3/2} \leq 2 \iff 2^3 \leq 2^2$);
- `pow_two_gt_two` : $\forall q : \mathbb{N}, 1 < q \rightarrow 2 < 2^q$, the general arithmetic fact behind (4);
- `conjecture_00000004007_false`, collecting the two instances and the general fact; the accompanying comment records that homogeneity reduces the conjecture to exactly these false inequalities.

`lean4/Check.lean` runs `#print axioms` for each theorem: the `decide`-based theorems depend on no axioms, and `pow_two_gt_two` and the final theorem depend only on the core axioms `propext` and `Quot.sound`; there is no `sorryAx` and no `ofReduceBool`. The real-valued p -variation itself, the homogeneity Lemma 3 and the reduction of the conjecture to (4) are proved here and checked with exact rational arithmetic in `reproduce.py`; they are not formalised in Lean (core Lean has no $\mathbb{R}$). See `lean4/README.md` for the precise scope.